

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS EXTENSION 2

HSC ASSESSMENT TASK

JUNE 2010

General Instructions:

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions.

NAME: _____

TEACHER: _____

Question 1	Question 2	Question 3	Total

Question 1 (15 marks)

Marks

- a) Find $\int \frac{1}{e^x + e^{-x}} dx$ 2
- b) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x}$ 3
- c) Find $\int \frac{\sin^{-1} x}{\sqrt{1+x}} dx$ 3
- d) i) Find real numbers a and b such that

$$\frac{5-3x}{(x+1)(x^2+1)} = \frac{a}{x+1} + \frac{bx+c}{x^2+1}$$
 3
 ii) Hence find $\int \frac{5-3x}{(x+1)(x^2+1)} dx$ 2
- e) Find $\int \operatorname{cosec} x \, dx$ 2

Question 2 (14 marks)

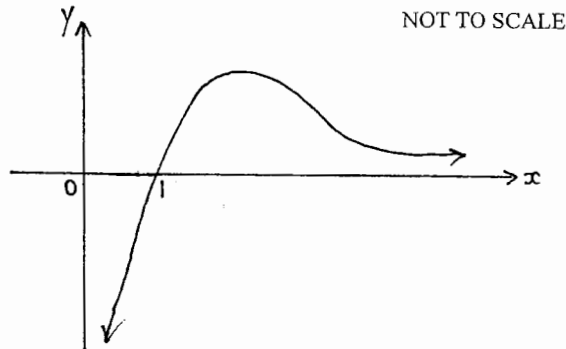
- a) The points $P(2t, \frac{2}{t})$ and $Q(2s, \frac{2}{s})$ lie on the hyperbola $xy = 4$.
 $(t \neq 0, s \neq 0, t^2 \neq s^2)$.
- i) Prove that the equation of the tangent to the hyperbola at the point P is
 $x + t^2 y = 4t$ 2
- ii) Prove that the tangents at P and Q intersect at
 $M(\frac{4st}{s+t}, \frac{4}{s+t})$ 2
- iii) Suppose that $s = \frac{-1}{t}$. Prove that the locus of M is a straight line and state any restrictions that may apply. 2
- b) Sketch without using calculus showing all important features:
 $y = \sin^{-1}(\sin x)$ $D: -\pi \leq x \leq \pi$ 2

Marks

- c) The equation $x^3 + x^2 - 2x + 1 = 0$ has roots α, β, γ .
- Show that α, β, γ are not integers. 2
 - Find the monic equation with roots $\alpha + 1, \beta + 1, \gamma + 1$. 2
 - Hence using both polynomials above or otherwise find the value of $(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma)$. 2

Question 3 (15 marks)

- a) Consider the polynomial $P(x) = x^4 + 2x^3 + x^2 - 1$.
It is given that one zero is $\frac{-1+i\sqrt{3}}{2}$. Find the other 3 zeros. 3
- b) The curve $y = f(x) = \frac{\log_e x}{x}$ is shown below.



Given the maximum turning point is $(e, \frac{1}{e})$, sketch the following curves showing essential features, using at least $\frac{1}{3}$ page for each.

- $y = f(x + 1)$ 1
 - $y = f(|x|)$ 1
 - $y = \frac{1}{f(x)}$ 3
- c) i) Given $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$, deduce $8x^3 - 6x - 1 = 0$ has solutions $x = \cos \theta$ where $\cos 3\theta = \frac{1}{2}$. 2
- ii) Find the roots of $8x^3 - 6x - 1 = 0$ in the form $\cos \theta$. 3
- iii) Hence evaluate $\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$. 2

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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Solutions 2010 Ext. 2 Task 2

Question 1

$$a) \int \frac{1}{e^x + e^{-x}} dx$$

$$\int \frac{e^{-x}}{(e^x)^2 + 1} dx$$

$$\text{Let } v = e^x$$

$$dv = e^x dx$$

$$\int \frac{dv}{v^2 + 1} \quad (1)$$

$$= \tan^{-1}(e^x) + C \quad (1)$$

$$b) \int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x}$$

$$\int \frac{\frac{2dt}{1+t^2}}{1 + \frac{1+t^2}{1+t^2} + \frac{2t}{1+t^2}} \quad (1)$$

$$\int \frac{2dt}{1+t^2 + 1 - t^2 + 2t}$$

$$\int \frac{dt}{1+t}$$

$$= \left[\log_e(1 + \tan \frac{x}{2}) \right]_0^{\frac{\pi}{2}} \quad (1)$$

$$= \log_e(1+1) - \log_e 1$$

$$= \log_e 2 \quad (1)$$

$$c) \int \frac{\sin^{-1} x}{\sqrt{1+x}} dx$$

$$\int \sin^{-1} x \times \frac{d}{dx}(2\sqrt{1+x}) dx \quad (1)$$

$$= \sin^{-1} x \times 2\sqrt{1+x} - \int \sqrt{1-x^2} \times 2\sqrt{1+x} dx \quad (1)$$

$$= 2\sqrt{1+x} \sin^{-1} x - \int \frac{2}{\sqrt{1-x}} dx$$

$$= 2\sqrt{1+x} \sin^{-1} x + 4\sqrt{1-x} + C \quad (1)$$

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$$d) i) \frac{5-3x}{(x+1)(x^2+1)} = \frac{a}{x+1} + \frac{bx+c}{x^2+1}$$

$$5-3x = a(x^2+1) + (bx+c)(x+1)$$

$$= ax^2 + a + bx^2 + bx + cx + c$$

$$-3x + 5 = (a+b)x^2 + (b+c)x + (a+c)$$

$$a+b=0$$

$$b+c=-3$$

$$a+c=5$$

$$a=-b$$

$$\therefore b+c=-3$$

$$-b+c=5$$

$$2c=2$$

$$c=1 \quad (1)$$

$$\therefore b=-4$$

$$\therefore a=4 \quad (1)$$

$$ii) \int \frac{4}{x+1} + \frac{-4x+1}{x^2+1} dx \quad (1)$$

$$= 4 \log_e |x+1| - 2 \log_e(x^2+1) + \tan^{-1} x + C \quad (1)$$

$$e) \int \operatorname{cosec} x \cdot dx$$

$$= \int \frac{\operatorname{cosec} x (\operatorname{cosec} x - \cot x)}{\operatorname{cosec} x - \cot x} d\theta = \log_e |\operatorname{cosec} x - \cot x| \quad (1)$$

Question 2

$$a) i) xy=4$$

$$y = 4x^{-1}$$

$$y' = -\frac{4}{x^2}$$

$$\therefore \text{At } x=2+$$

$$y' = -\frac{4}{4+^2}$$

$$m_{\text{tangent}} = -\frac{1}{4+^2}$$

$$y - \frac{-1}{4+^2} = -\frac{1}{4+^2}(x-2+)$$

$$+^2 y - 2+ = -x + 2+$$

$$x + + y = 4+$$

$$ii) \text{ Tangents at P and Q are}$$

$$x + +^2 y = 4+$$

$$x + s^2 y = 4s \quad \text{Simultaneous}$$

$$y(+^2 - s^2) = 4(+ - s)$$

$$y = \frac{4}{s + +}$$

$$\therefore x = 4+ - +^2 \times \frac{4}{s + +}$$

$$= \frac{4+s + 4+^2 - 4+^2}{s + +}$$

$$= \frac{4s +}{s + +}$$

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So M is

$$\left(\frac{4s+t}{s+t}, \frac{4}{s+t} \right)$$

iii) If $s+t = -1$

$$x = \frac{-4}{s+t}, y = \frac{4}{s+t}$$

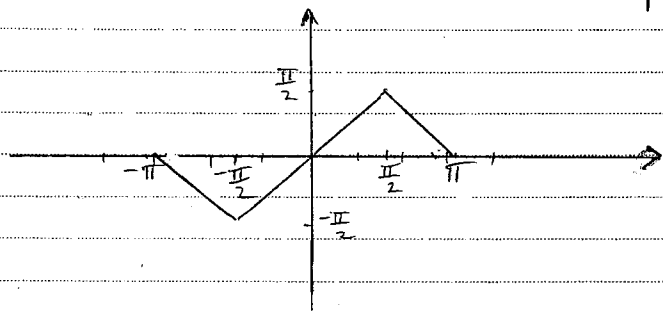
$\therefore x = -y$ is locus of M but

$s \neq 0$ and $t \neq 0$

$\therefore (0,0)$ is not part of locus.

b) $y = \sin^{-1}(\sin x)$ $D: -\pi \leq x \leq \pi$
has range the same as $y = \sin^{-1}x$
ie: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$. It is also odd

since $\sin x$ is odd. Also period is 2π



a) i) $x^3 + x^2 - 2x + 1 = 0$ has roots α, β, γ .
By Factor Theorem roots must be factors of 1 ie: 1 or -1 if they are integers.
 $P(1) = 1^3 + 1^2 - 2(1) + 1 = 0$
 $P(-1) = (-1)^3 + (-1)^2 - 2(-1) + 1 = 0$
 $\therefore$ Roots are not integers.

ii) Replace x with $x-1$ in original:

$$(x-1)^3 + (x-1)^2 - 2(x-1) + 1 = 0$$

$$x^3 - 3x^2 + 3x - 1 + x^2 - 2x + 1 - 2x + 2 + 1 = 0$$

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$$x^3 - 2x^2 - x + 3 = 0 \text{ is monic equation.}$$

ii) From original polynomial,

$$\alpha + \beta + \gamma = -1$$

$$\therefore \alpha + \beta = -1 - \gamma$$

$$\alpha + \gamma = -1 - \beta$$

$$\beta + \gamma = -1 - \alpha$$

$$\therefore (\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) = -(\alpha + 1)(\beta + 1)(\gamma + 1) = -(\alpha + 1)(\beta + 1)(\gamma + 1)$$

which is the negative of the product of the roots in $x^3 - 2x^2 - x + 3 = 0$

$$\text{ie: } -1 \times \frac{-3}{1} = 3.$$

Question 3

a) $P(x) = x^4 + 2x^3 + x^2 - 1$

One zero is

$$\frac{-1 + i\sqrt{3}}{2} \therefore \text{another root is } \frac{-1 - i\sqrt{3}}{2}$$

as polynomial has real coefficients.

This quadratic factor must be:

$$x^2 - \left(\frac{-1 + i\sqrt{3}}{2} + \frac{-1 - i\sqrt{3}}{2} \right)x + \frac{1 + 3}{4} = 0$$

$$x^2 + x + 1 = 0$$

$\therefore$ By inspection:

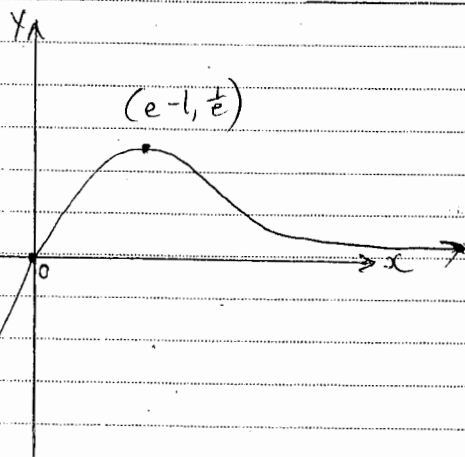
$$x^4 + 2x^3 + x^2 - 1 = (x^2 + x + 1)(x^2 + x - 1)$$

$$\text{Other zeros are } \frac{-1 \pm \sqrt{1 - 4(1)(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2}$$

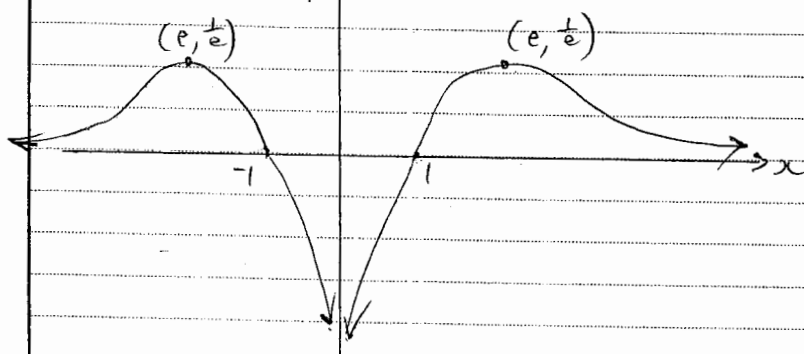
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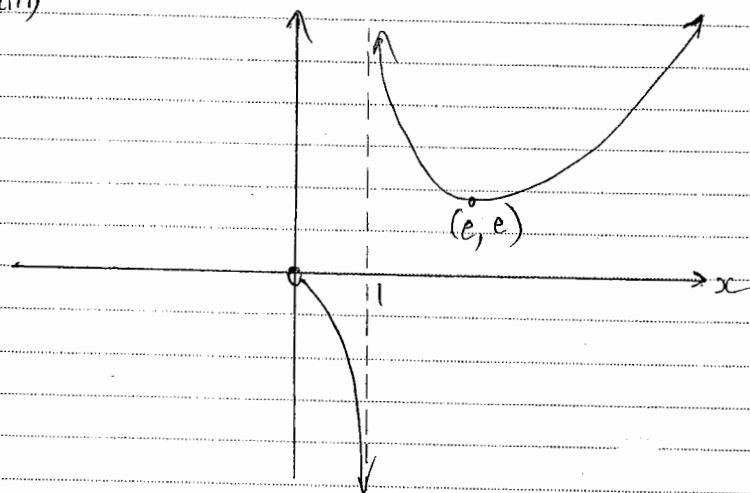
b) ci)



ci)



cii)



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$$c) \text{ i) If } \cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

$$2\cos 3\theta = 8\cos^3\theta - 6\cos\theta$$

$$2\cos 3\theta - 1 = 8\cos^3\theta - 6\cos\theta - 1$$

So in $8x^3 - 6x - 1 = 0$ if $x = \cos\theta$, solutions for θ are the same as for

$$2\cos 3\theta - 1 = 0$$

$$2\cos 3\theta = 1$$

$$\cos 3\theta = \frac{1}{2}$$

$$\text{ii) } \cos 3\theta = \frac{1}{2}$$

$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}, \frac{17\pi}{3}$$

$$\therefore \theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}, \frac{17\pi}{9}$$

$\therefore$ Solutions are $\cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}, \cos \frac{11\pi}{9}, \cos \frac{13\pi}{9}, \cos \frac{17\pi}{9}$
S A M E

$\therefore$ Roots are $\cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}$

$$\text{iii) } \cos \frac{5\pi}{9} = -\cos \frac{4\pi}{9}, \quad \cos \frac{7\pi}{9} = -\cos \frac{2\pi}{9}$$

$\therefore \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$ is the product of the roots in this equation and is calculated by $-\frac{d}{a}$

$$\therefore \frac{1}{8}$$